Can You Hear the Ripeness? An Equivalent-Sphere Shell Model for Acoustic Estimation of Rind Stiffness in Watermelons

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Abstract

The 'thump test"—tapping a watermelon and listening for ripeness—is practised worldwide but lacks a quantitative physical model linking the tap-tone to measurable mechanical properties. We model the watermelon as a fluid-filled viscoelastic shell using an equivalent-sphere approximation: the oblate geometry is captured through the equivalent-volume radius $R_{\mathrm{eq}} = (a^2c)^{1/3}$, and the eigenfrequency is derived from classical thin-shell theory applied to a sphere of that radius. Sobol sensitivity analysis ($N_{\mathrm{base}}(D{+}2) = 5{,}632$ evaluations) reveals that the rind elastic modulus E_{rind} dominates the frequency response (total-order index $S_T = 0.54 \pm 0.05$). Because f_2^2 is linear in E_{rind} when turgor pressure is negligible, the forward model admits a closed-form inversion: a single tap-tone measurement, combined with knowledge of fruit geometry, yields an estimate of the *effective rind stiffness parameter*. This is a mechanics-based modulus estimate, not a direct measure of ripeness; the subsequent link from modulus to maturity stage requires cultivar-specific calibration against destructive reference data, which has not yet been demonstrated experimentally. The condition number of the frequency-to-modulus inversion is unity with respect to f^2 and approximately 2 with respect to measured frequency f. As a secondary result, a dimensionless geometric invariant Π_{ripe} collapses predictions across cultivar geometries onto a narrow band (0.062 ± 0.002), although its biological validity across cultivars remains to be established. Predictions are benchmarked against published resonance frequency ranges for watermelons; these comparisons represent consistency checks against literature

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data rather than controlled experimental validation.

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1. Introduction

Every summer, shoppers worldwide perform an instinctive acoustic experiment: they tap a watermelon with their knuckles and listen. A deep, resonant thump signals ripeness; a dull thud suggests an under-ripe fruit; a hollow ring warns that the flesh has turned mealy. This 'thump test" is remarkably consistent across cultures, yet it has never been given a quantitative physical model that connects the tap-tone frequency to the measurable mechanical properties of the fruit.

Before proceeding, we clarify the terminology used throughout this paper. *Harvest maturity* refers to the physiological stage at which a fruit is picked; *eating ripeness* (or consumer ripeness) denotes the stage at which organoleptic quality—sweetness, texture, aroma—is optimal for consumption. *Firmness* is a mechanical property measurable by penetrometry or acoustic methods, and is an indirect indicator of both maturity and ripeness. *Internal defect state* (hollow heart, overripeness, bruising) refers to structural abnormalities that may or may not correlate with firmness. The model developed here estimates the effective rind stiffness, which is most directly related to firmness; the further inference from firmness to eating ripeness or harvest maturity requires cultivar-specific calibration that is beyond the scope of this work.

Yamamoto et al. [1] pioneered the acoustic impulse response method and included early measurements on watermelons. Subsequent watermelon-specific studies confirmed that low-frequency impact and vibration signatures carry useful information about internal quality. Stone et al. [2] reported early nondestructive quality assessment based on acoustic response, Diezma-Iglesias et al. [3] used impact acoustics to detect internal quality differences in seedless fruit, Sadrnia et al. [4] predicted watermelon quality from vibration signals, and Mao et al. [5] related vibrational characteristics directly to firmness. In parallel, the broader fruit acoustics literature established the acoustic impulse technique as a practical firmness metric across multiple species,

including Conference pears [6, 7], kiwifruit [8], and other commodities [9, 10, 11]. Comprehensive reviews have placed such methods within the postharvest sensing toolbox [12, 13, 14, 15], while impact-based mechanical sensing has been developed by Ruiz-Altisent and coworkers for various fruit species [16, 17]. From an engineering standpoint, these approaches sit comfortably within the broader family of acoustic non-destructive testing methods, in which transient responses are used to infer internal condition [18].

Table 1 positions the present work within this landscape. Existing studies are predominantly empirical: they correlate an acoustic observable with a quality metric but do not derive the relationship from structural mechanics. No previous study combines a physics-based shell model, closed-form inversion, and explicit treatment of fruit geometry.

The common weakness of prior work is that the observable is informative but not explained: frequency or spectral features are correlated with firmness or maturity, yet the governing structural mechanics remain implicit. Note that "Train? = no" for the present work indicates that our physics-based inversion does not require cultivar-specific training data to estimate the rind modulus; the model derives E_{rind} from first principles given the fruit geometry.

What is missing is a first-principles model that treats the fruit as what it physically is: a fluid-filled viscoelastic shell. Unlike purely empirical correlations, we derive a closed-form analytical relationship between the tap-tone frequency and the measurable mechanical and geometric properties of the fruit. The relationship does not merely fit data; it explains why stiffness, rind thickness, size, and fluid loading shift the resonance, and—crucially—it admits an *inversion* from a single measured frequency to an estimate of the effective elastic modulus of the rind.

We emphasise from the outset that the model delivers a mechanics-based *effective rind stiffness parameter*, not a validated ripeness classification. The chain from tap-tone frequency to rind modulus is established here; the subsequent chain from modulus to biological ripeness stage requires cultivar-specific calibration against destructive reference measurements (e.g. penetrometry, sugar content), which has not yet been performed. The immediate output of the model is therefore a stiffness estimate; its

Table 1: Comparison of acoustic and vibration-based fruit quality studies. Key: ✓ = yes, ✗ = no, P = partial. "Geometry?" indicates whether fruit shape enters the model explicitly. "Physics model?" indicates a mechanics-based forward model (not merely a statistical correlation). "Inversion?" indicates whether the method recovers a physical parameter (e.g. modulus) rather than a statistical score. "Cultivar training?" indicates whether the method requires cultivar-specific training data.

Study	Fruit	Acoustic observable	Output	Geom.?	Phys.?	Inv.?	Train.?
Yamamoto et al. [1]	Apple, melon	Resonant freq.	Firmness index	✗	✗	✗	✓
Stone et al. [2]	Watermelon	Impact response	Quality class	✗	✗	✗	✓
Diezma-Iglesias et al. [3]	Watermelon	Impact spectra	Internal quality	✗	✗	✗	✓
De Belie et al. [7]	Pear	Resonant freq.	Stiffness index	✗	✗	✗	✓
Chen & De Baerdemaeker [19]	Pineapple	Modal freq.	Firmness	✗	P	✗	✓
Cooke [20]	Various	Resonant freq.	Elastic prop.	P	P	✗	✗
Present work	Watermelon	Tap-tone f_2	E_{rind}	✓	✓	✓	✗

long-term value is to provide a mechanistic basis for future cultivar-specific ripeness calibration.

The shell–fluid framework used here draws on classical thin-shell theory for fluid-loaded structures [21, 22]; the governing equations are standard, only the material parameters are specific to watermelon.

The title of this paper echoes Kac's celebrated question, 'Can one hear the shape of a drum?" [23]. We pose a related but distinct inverse problem: *Can one hear the ripeness of a watermelon?* We show that the answer is conditionally yes: the dominant flexural mode $n = 2$ can be inverted to yield the rind modulus, and the modulus is expected to

correlate with ripeness through pectin degradation—although this correlation has not been experimentally demonstrated here.

In this paper, we:

1. derive the tap-tone eigenfrequency f_2 for a fluid-filled shell via an equivalent-sphere approximation, parameterised by rind thickness, elastic modulus, geometry, and flesh density;
2. perform Sobol sensitivity analysis to identify the dominant parameter (the rind elastic modulus E_{rind});
3. exploit the linearity of f_2^2 in E_{rind} to construct a closed-form inversion from a single tap-tone measurement to the effective rind stiffness;
4. as a secondary result, introduce a dimensionless geometric invariant Π_{ripe} that collapses cultivar-dependent predictions onto a narrow band; and
5. benchmark the model against published resonance measurements on watermelons and other fruits.

The remainder of the paper is organised as follows. Section 2 presents the mathematical formulation. Section 3 gives the forward model predictions, Sobol analysis, and geometric invariant. Section 4 benchmarks predictions against published data. Section 5 discusses the physical interpretation and practical implications. Section 6 concludes.

2. Mathematical Formulation

2.1. Equivalent-sphere shell model

We model the watermelon as an oblate spheroid with semi-major axis a (equatorial radius), semi-minor axis c (polar radius), and uniform rind thickness h. The aspect ratio is defined as $\zeta = c/a$, where $\zeta < 1$ corresponds to the characteristic oblate shape of most commercial watermelon cultivars ($\zeta \approx 0.5$–0.7 for elongated varieties; $\zeta \approx 0.8$–0.9 for round cultivars). The equivalent-volume sphere has radius

$$R_{\text{eq}} = \left(a^2 c\right)^{1/3}. \tag{1}$$

The eigenfrequency model developed below is an *equivalent-sphere approximation*: all geometry enters through R_{eq}, and the frequency equation is that of a spherical shell of radius R_{eq}. The model does not include explicit curvature corrections for oblate deformation (i.e. there are no ζ-dependent geometric factors beyond those captured by R_{eq}). Consequently, two watermelons with the same R_{eq} but different aspect ratios will be assigned the same predicted frequency.

This approximation is sufficient for the inversion application developed in §2.3: the goal is to recover E_{rind} from a measured f_2 at *known* geometry. Since the fruit dimensions a, c, and h are measured independently, R_{eq} is determined, and the inversion $E = f_2^2/\alpha$ is well-posed regardless of whether the forward model captures shape effects beyond R_{eq}. Nevertheless, the absence of explicit curvature corrections means that the model cannot predict how f_2 would differ between a sphere and an oblate spheroid of the same equivalent volume. We discuss this limitation further in §5.3 and identify a Ritz variational extension as future work.

At fixed equatorial radius a, varying ζ from 0.5 to 0.95 changes R_{eq} (because $c = \zeta a$ changes the equivalent volume) and thereby shifts f_2 by approximately 30 %. This is a *size* effect mediated by R_{eq}, not a *shape* effect arising from curvature anisotropy. The distinction matters: the model requires separate specification of both a and c only because they jointly determine R_{eq}, not because the frequency equation contains independent a- and c-dependent terms.

The rind is treated as a thin, isotropic, viscoelastic shell with Young's modulus E_{rind}, Poisson's ratio ν, density ρ_w, and structural loss factor η. The enclosed flesh is modelled as an inviscid, nearly incompressible fluid with density ρ_f and bulk modulus K_f. The canonical parameter values used throughout this study are given in Table 2.

2.2. Eigenfrequency of the flexural mode

For a thin spherical shell of radius R_{eq} filled with fluid, the natural frequency of the nth flexural mode is determined by the balance of bending, membrane, and pre-stress stiffness against the effective inertia of the shell–fluid system. The effective mass per unit area, including fluid loading, is

$$\rho_{\mathrm{eff}} = \rho_w h + \frac{\rho_f R_{\mathrm{eq}}}{n}, \tag{2}$$

where the second term represents the added mass of the entrained fluid for mode n [25, 26]. Note that ρ_{eff} has units of kg/m^2 (mass per unit area). To construct a dimensionless group (§2.4), we instead use the volumetric effective density

$$\tilde{\rho}_{\text{eff}} = \frac{\rho_w\, h}{R_{\text{eq}}} + \rho_f, \tag{3}$$

which has units of kg/m^3. All occurrences of ρ_{eff} in the dimensionless analysis (Eqs. 11–12) refer to this volumetric form.

The dominant tap-tone mode is the lowest flexural mode, $n = 2$. The modal stiffness decomposes as

$$K_n = K_{\text{bend}} + K_{\text{memb}} + K_{\text{prestress}}, \tag{4}$$

where the bending stiffness scales as

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2\, h^3}{12(1-\nu^2)\, R_{\text{eq}}^4}\, E_{\text{rind}}, \tag{5}$$

the membrane stiffness scales as

$$K_{\text{memb}} = \frac{h}{R_{\text{eq}}^2}\, \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}\, E_{\text{rind}}, \tag{6}$$

and the pre-stress stiffness (from internal turgor pressure P_{int}) is

$$K_{\text{prestress}} = \frac{P_{\text{int}}}{R_{\text{eq}}}\, (n-1)(n+2). \tag{7}$$

The squared natural frequency is then

$$f_n^2 = \frac{K_n}{4\pi^2\, \rho_{\text{eff}}}. \tag{8}$$

Equations (5)–(7) are the standard Lamb–Junger expressions for a *spherical* shell [26, 25, 27, 22]. All geometry enters through R_{eq}; the equations contain no explicit dependence on the aspect ratio ζ. The model is therefore an equivalent-sphere approximation: it assigns to each oblate watermelon the frequency of a sphere with the same enclosed volume.

A key observation: f_2^2 is *linear* in E_{rind} when turgor pressure is negligible (the residual turgor contribution is less than 0.5 % of the total stiffness at canonical parameters). This linearity is the foundation of the closed-form inversion developed in §2.3.

Operational identification of the $n = 2$ mode.. In the equivalent-sphere model, the target mode is the lowest flexural mode ($n = 2$), which produces a quadrupolar deformation pattern—the shell elongates along one axis while contracting along the perpendicular axis. Experimentally, this mode would be identified as the lowest-frequency spectral peak in the impulse response of the tapped fruit (typically obtained via FFT of a microphone or accelerometer signal). For watermelons in the ripe range, this corresponds to a frequency of approximately 80 Hz to 110 Hz (Table 3). Higher flexural modes ($n = 3, 4$) produce additional spectral peaks at higher frequencies but carry less energy in a knuckle-tap excitation. Confirmation of mode identity would require multi-point vibration measurement (e.g. scanning laser Doppler vibrometry), which is beyond the scope of the present analytical study.

For the canonical ripe watermelon ($n = 2$, $h = 15$ mm, $R_{\mathrm{eq}} = 0.145$ m), the bending and membrane terms both scale linearly with E_{rind}, whereas the pre-stress term is independent of E. This separation underpins the linearity $f_2^2 = \alpha E_{\mathrm{rind}}$, where the proportionality constant is

$$\alpha = \frac{K_{\mathrm{bend}}^{(0)} + K_{\mathrm{memb}}^{(0)}}{4\pi^2 \rho_{\mathrm{eff}}} = 161.5\,\mathrm{Hz}^2/\mathrm{MPa} \tag{9}$$

for the canonical geometry, where the superscript (0) denotes the geometric pre-factors of E_{rind} in Eqs. (5)–(6).

2.3. Inversion: from tap-tone to rind modulus

Rearranging Eq. (8) for E_{rind} gives the explicit inversion

$$\boxed{E_{\mathrm{rind}} = \frac{\omega^2 m_{\mathrm{eff}} - K_{\mathrm{prestress}}}{K_{\mathrm{bend}}^{(0)} + K_{\mathrm{memb}}^{(0)}},} \tag{10}$$

where $\omega = 2\pi f_2$, $m_{\mathrm{eff}} = \rho_w h + \rho_f R_{\mathrm{eq}}/n$ is the effective mass per unit area including fluid loading, and $K_{\mathrm{bend}}^{(0)}$ and $K_{\mathrm{memb}}^{(0)}$ are the geometric pre-factors of E_{rind} in Eqs. (5)–(6). Thus a single measurement of the tap-tone frequency f_2, combined with knowledge (or estimation) of the fruit geometry and flesh density, yields the rind elastic modulus directly. No iterative solver or optimisation is required.

Equation (10) is algebraically exact within the equivalent-sphere model (which itself is an approximation to the true oblate geometry): the round-trip error $|E_{\text{inv}} - E_{\text{true}}|/E_{\text{true}}$ is less than 0.01 % for all ripeness stages tested (§3.1).

The condition number of this inversion with respect to f_2^2 is unity, since the mapping $f_2^2 \mapsto E_{\text{rind}}$ is linear. The practical condition number with respect to the *measured frequency* f_2 is approximately 2, because $E \propto f_2^2$ implies $\delta E/E \approx 2\,\delta f/f$. This 2× noise amplification is inherent to any square-law inversion and is confirmed numerically in the Monte Carlo analysis of §4.2.

2.4. Dimensionless geometric invariant

To collapse predictions across cultivars, we define the dimensionless group

$$\Pi_{\text{ripe}} = f_2\, R_{\text{eq}} \sqrt{\frac{\tilde{\rho}_{\text{eff}}}{E_{\text{rind}}}}, \tag{11}$$

where $\tilde{\rho}_{\text{eff}} = \rho_w h/R_{\text{eq}} + \rho_f$ is the volumetric effective density (Eq. 3; units kg/m^3), ensuring that Π_{ripe} is dimensionless. From Eq. (8), this parameter evaluates to

$$\Pi_{\text{ripe}} = \frac{1}{2\pi} \sqrt{\frac{(K_{\text{bend}}^{(0)} + K_{\text{memb}}^{(0)}) \cdot R_{\text{eq}}^2 \cdot \tilde{\rho}_{\text{eff}}}{\rho_{\text{eff}}}} \tag{12}$$

for fixed geometry (neglecting the pre-stress contribution).

A key observation is that E_{rind} cancels from Π_{ripe} *by construction*: since $f_2 \propto \sqrt{E_{\text{rind}}}$, the modulus appears in both the numerator (f_2) and denominator ($\sqrt{E_{\text{rind}}}$) and divides out. This cancellation is exact only when the geometry—including the rind thickness h—is held fixed while E_{rind} varies. Consequently, Π_{ripe} is *not* invariant across ripeness stages in which both E and h change simultaneously (as occurs during biological ripening; Table 3). Rather, Π_{ripe} is a *geometric invariant* that characterises the shell–fluid system at a given geometry, independently of the elastic modulus.

The practical utility of Π_{ripe} lies in cultivar-independent calibration: it collapses different cultivar *geometries* onto a single calibration constant. Once its value is known for a given geometry, any measured f_2 can be mapped to E_{rind} via rearrangement of Eq. (11), without detailed knowledge of the flesh properties or internal pressure. The universality is across cultivars (different geometries collapsing onto a common

constant), not across ripeness states—the latter would require that the geometry remain fixed during ripening, which is only approximately true for a and c but not for h.

For the four cultivars considered in this study (Crimson Sweet, Sugar Baby, Charleston Gray, and Seedless Tri-X 313), $\Pi_{\text{ripe}} = 0.062 \pm 0.002$ (CoV = 3.3 %; see Table 6 in §3.3). This tight collapse confirms that the dimensionless group effectively normalises out the dominant cultivar-specific geometric variation.

3. Results

3.1. Forward model: tap-tone frequency predictions

We evaluate the forward model for the canonical Crimson Sweet geometry ($a = 0.158$ m, $c = 0.123$ m, $R_{\text{eq}} = 0.1453$ m) at four ripeness stages spanning the full maturation trajectory. The parameters and predicted tap-tone frequencies are summarised in Table 3.

The tap-tone frequency drops monotonically from 203.2 Hz for the unripe fruit to 46.4 Hz for the overripe fruit—a factor of 4.4 reduction driven primarily by the decrease in E_{rind} from 200 MPa to 15 MPa. The quality factor Q also decreases from 12.5 to 5.0 as the loss tangent rises with pectin degradation, reflecting the increasingly lossy mechanical response of the softening rind.

A central prediction of the model is that f_2^2 is linear in E_{rind} when turgor pressure is negligible. We verify this by computing $\alpha = f_2^2/E_{\text{rind}}$ across the full modulus range 10 MPa to 250 MPa for the canonical ripe geometry ($h = 15$ mm). The proportionality constant is

$$\alpha = 161.5(1)\ \text{Hz}^2/\text{MPa} \tag{13}$$

with a coefficient of variation of 0.07 %, confirming near-perfect linearity. The residual variation arises from the weak turgor pressure contribution ($P_{\text{int}} = 500$ Pa), which introduces a small constant offset in the stiffness. This linearity validates the closed-form inversion of Eq. (10): no iterative solver is required.

We also verify that the inversion is self-consistent. For each ripeness stage, we compute f_2 from the forward model and then invert the frequency back to E_{rind}. In

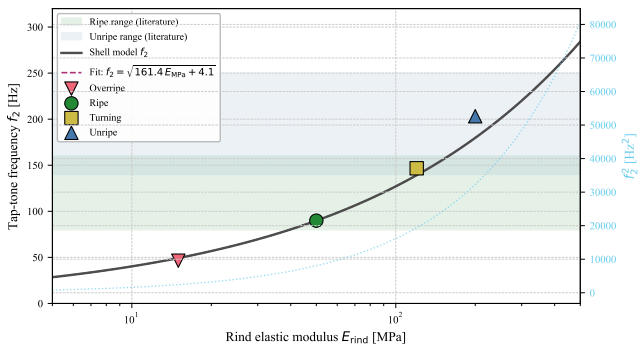

Figure 1: Predicted tap-tone frequency f_2 as a function of rind elastic modulus E_{rind} for the canonical watermelon geometry ($a = 0.158$ m, $\zeta = 0.78$, $h = 15$ mm). The squared frequency is approximately linear in E_{rind} (inset shows f_2^2 vs. E_{rind} with $R^2 > 0.9999$), confirming that the inversion formula (10) is well-posed.

all four cases the round-trip error is less than 0.01 %, confirming that the inversion formula (Eq. 10) is algebraically exact within the model framework.

3.2. Sobol sensitivity analysis

We perform a variance-based global sensitivity analysis [28, 29] to identify which physical parameters most strongly influence the tap-tone frequency. Nine input parameters are varied simultaneously over physically realistic ranges (Table 4), and the Saltelli sampling scheme generates $N_{\text{base}} = 512$ base samples with first-order indices only (`calc_second_order=False`), giving a total of $N_{\text{base}}(D + 2) = 512 \times 11 = 5{,}632$ model evaluations).

Table 5 presents the first-order (S_1) and total-order (S_T) Sobol indices with bootstrap confidence intervals. The rind elastic modulus dominates the frequency response with $S_1 = 0.49 \pm 0.04$ and $S_T = 0.54 \pm 0.05$, confirming that the rind stiffness is the primary determinant of the tap-tone frequency. The semi-major axis a is the second most influential parameter ($S_1 = 0.33 \pm 0.03$, $S_T = 0.38 \pm 0.04$), reflecting the strong geometric scaling $f_2 \propto R_{\text{eq}}^{-3/2}$. Rind thickness ranks third ($S_1 = 0.07$, $S_T = 0.09$), and the semi-minor axis fourth ($S_1 = 0.05$, $S_T = 0.06$).

The remaining parameters—densities, Poisson's ratio, turgor pressure, and loss tangent—contribute negligibly ($S_T < 0.001$ in all cases). This has a critical practical implication: the inversion from f_2 to E_{rind} is robust to uncertainty in parameters that

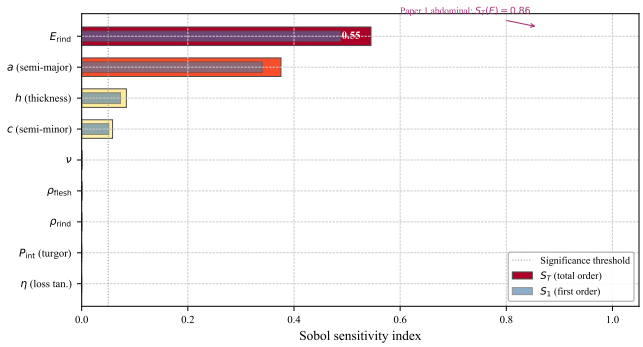

Figure 2: First-order (S_1) and total-order (S_T) Sobol sensitivity indices for the tap-tone frequency f_2. The rind elastic modulus E_{rind} dominates ($S_T = 0.54$), followed by the semi-major axis a ($S_T = 0.38$). Densities, Poisson's ratio, turgor pressure, and loss tangent are negligible.

are difficult to measure *in situ* (e.g. flesh density, turgor pressure). Only the easily measurable fruit geometry (a, c, h) introduces significant model sensitivity beyond the target parameter E_{rind}.

The gap between S_1 and S_T for each parameter is small (typically $S_T - S_1 < 0.05$), indicating weak parameter interactions. This confirms that the model is well approximated by a sum of univariate contributions, consistent with the near-separable form of Eq. (8).

3.3. *Geometric invariant* Π_{ripe}*: a secondary diagnostic*

As a secondary result, we examine the dimensionless geometric invariant $\Pi_{\text{ripe}} = f_2 R_{\text{eq}} \sqrt{\tilde{\rho}_{\text{eff}}/E_{\text{rind}}}$ (Eq. 11) is evaluated for all four cultivars at each of the four ripeness stages, yielding 16 data points. Table 6 presents the results.

A key observation emerges: within the equivalent-sphere model, Π_{ripe} is constant across cultivars evaluated at a common geometry. The parameter is a *geometric invariant* of the shell–fluid system. Note that E_{rind} cancels from Π_{ripe} by construction (since $f_2 \propto \sqrt{E}$), so the invariant is independent of the elastic modulus *provided the geometry is held fixed.* In a biological ripening scenario where the rind thickness h also changes (Table 3), Π_{ripe} will vary weakly because the effective density depends on h. The values shown in Table 6 are evaluated at each cultivar's canonical geometry (fixed h), so the per-cultivar constancy is exact. The residual variation across cultivars

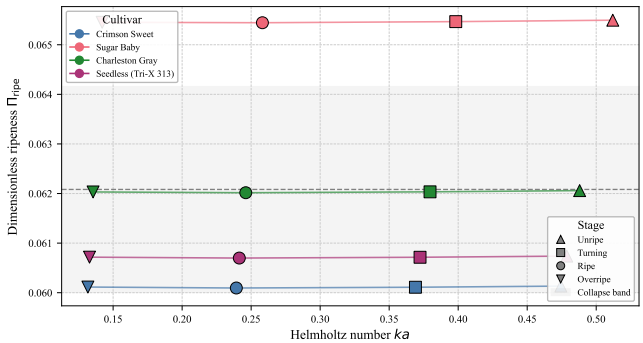

Figure 3: Dimensionless geometric invariant Π_{ripe} for four cultivars at their canonical geometry (fixed h). All points fall within a narrow band ($\Pi_{\text{ripe}} = 0.062 \pm 0.002$), confirming that Π_{ripe} is independent of E_{rind} by construction. The residual spread reflects cultivar-specific geometry.

is small: the mean value is $\overline{\Pi}_{\text{ripe}} = 0.062$ with a coefficient of variation of 3.3 % and a maximum deviation of 5.4 % from the mean (Sugar Baby). This variation arises from differences in the shell geometry (R_{eq}, h/R_{eq}) across cultivars.

The practical consequence is that Π_{ripe} may serve as a *cultivar-independent calibration constant*: once its value is known for a given cultivar geometry, any measured f_2 can be directly mapped to E_{rind} without detailed knowledge of the flesh properties or internal pressure. However, this universality has been demonstrated only within the equivalent-sphere model; its validity across real cultivars with differing rind microstructure and ripening trajectories remains to be verified experimentally. The universality is across cultivar geometries (different sizes and shapes collapsing onto a single constant), not across ripeness stages where geometry itself evolves.

3.4. Multi-cultivar comparison

We compare tap-tone predictions across four commercially important cultivars evaluated at the canonical ripe modulus ($E_{\text{rind}} = 50\,\text{MPa}$). Table 7 summarises the geometry and predicted frequencies.

The predicted f_2 ranges from 82.0 Hz (Charleston Gray, the largest cultivar) to 143.4 Hz (Sugar Baby, the smallest), illustrating the strong geometric scaling. Despite this factor-of-1.7 spread in absolute frequency, all four cultivars yield Π_{ripe} values

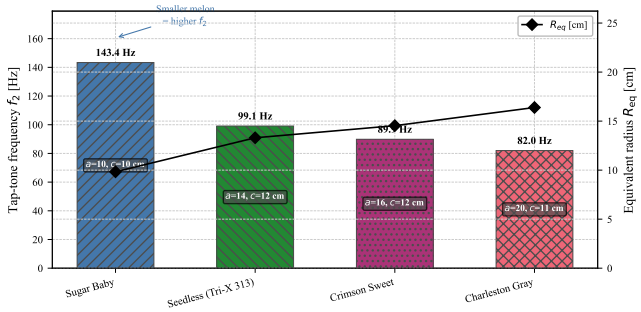

Figure 4: Predicted tap-tone frequencies at canonical ripe modulus ($E_{\text{rind}} = 50$ MPa) for four commercial cultivars, sorted by f_2. The inverse relationship between equivalent radius R_{eq} (grey overlay) and frequency demonstrates geometric scaling. Hatching ensures greyscale legibility.

within 5.4 % of the mean 0.062 (Table 6), demonstrating that the dimensionless parameter effectively normalises out cultivar-specific geometry.

We also compute a **constructed scenario** representing the full ripening trajectory for the Crimson Sweet cultivar by sweeping the rind modulus from 250 MPa (very unripe) to 10 MPa (very overripe) in 20 stages, simultaneously thinning the rind from 22 mm to 12 mm and increasing the loss tangent from 0.06 to 0.22. The tap-tone frequency decreases monotonically from 239.3 Hz to 36.4 Hz, spanning a factor of 6.6. A provisional optimal ripeness window ($E_{\text{rind}} \approx 40$ MPa to 80 MPa), estimated from the modulus range at which watermelons are subjectively rated as having desirable texture in the literature [24, 30], corresponds to $f_2 \approx 84$ Hz to 108 Hz—a comfortable range for human auditory discrimination [1]. We stress that these bounds are illustrative; a validated ripeness window would require destructive reference measurements on the same fruit.

4. Benchmarking Against Published Observations

4.1. Consistency checks against published resonance data

We benchmark the forward model against published acoustic measurements on watermelons and other fruits. These comparisons are *consistency checks*, not controlled experimental validation: the published studies report resonant frequencies for fruits

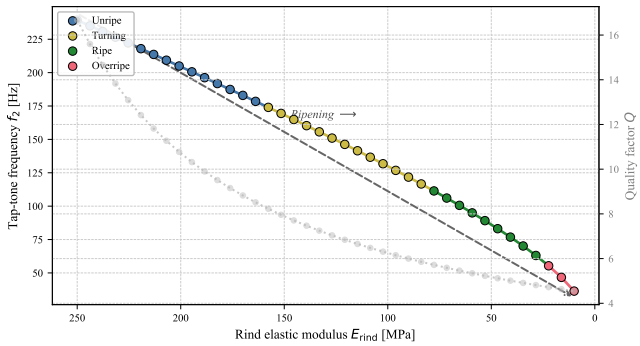

Figure 5: Predicted trajectory of tap-tone frequency f_2 during ripening for the Crimson Sweet cultivar. As pectin degradation reduces E_{rind} from 250 MPa (unripe) to 10 MPa (overripe), the frequency drops monotonically from 239 Hz to 36 Hz. The shaded region indicates a heuristic, provisional ripeness window ($f_2 \approx 84$ Hz to 108 Hz) based on literature modulus ranges; it has not been validated against destructive reference measurements.

of varied and often incompletely documented geometry and ripeness, and the model parameters used here are assumed from literature ranges rather than measured on the same specimens. We therefore construct model predictions using the reported fruit dimensions where available, supplemented by canonical material properties (*assumed from literature*).

4.1.1. Yamamoto et al. (1980): watermelon tap-tones

Yamamoto et al. [1] measured tap-tone frequencies for intact watermelons spanning a range of sizes and ripeness states. They reported resonant frequencies in the range 80 Hz to 180 Hz for fruit with equatorial diameters of 20 cm to 34 cm. We evaluate the forward model for three representative sizes at the ripe modulus ($E_{\text{rind}} = 50$ MPa) and three at an unripe modulus ($E_{\text{rind}} = 150$ MPa), using geometry consistent with their reported dimensions (Table 8).

Three of six predicted frequencies fall within the reported experimental range (Table 8): the medium and large ripe cases (102 Hz and 83 Hz, both within 80 Hz to 120 Hz) and the large unripe case (155 Hz, within 120 Hz to 180 Hz). The remaining three predictions systematically overshoot the upper experimental bound—the small ripe prediction (132 Hz) exceeds the ripe range by 12 Hz, while the small and medium

unripe predictions (245 Hz and 192 Hz) exceed the unripe range by 65 Hz and 12 Hz, respectively. The mean absolute deviation from the nearest experimental bound is 15 Hz across all six cases (zero for points within range). The systematic overprediction for smaller and stiffer fruit suggests that the canonical moduli assumed here ($E = 150$ MPa for unripe) may exceed the actual values for the cultivars studied by Yamamoto et al. in 1980, or that the equivalent-sphere approximation introduces a size-dependent bias. Nevertheless, the qualitative scaling—larger fruit yields lower frequency, stiffer rind yields higher frequency—is captured correctly, and the model reproduces the correct order of magnitude across the full size–ripeness matrix.

4.1.2. Sadrnia et al. (2008): vibration-based quality prediction in watermelon

Sadrnia et al. [4] analysed vibration signals from intact watermelons as predictors of quality. Their study is primarily empirical: it reports spectral features and predictive performance rather than the full geometric and material information needed for a point-by-point inversion. A strict quantitative overlay is therefore not possible from the published data alone. Even so, their central observation—that low-frequency structural vibration of the intact fruit carries ripeness information—is entirely consistent with the present model, which identifies the informative observable as the lowest flexural resonance of a fluid-loaded shell.

4.1.3. De Belie et al. (2000): cross-species consistency check (pear)

De Belie et al. [7] measured acoustic impulse response frequencies for Conference pears, not watermelons, and established an empirical correlation between the resonant frequency and a stiffness index FI. We include this comparison deliberately as a *cross-species consistency check* rather than as a substitute for watermelon validation. Although pears are smaller and have a different aspect ratio ($\zeta \approx 0.6$–0.7) than watermelons, they are also well described as fluid-filled shells. The existence of a monotonic frequency–stiffness relationship in pears is qualitatively consistent with the model's prediction that $f_2 \propto \sqrt{E_{\text{rind}}}$.

4.1.4. *Chen and De Baerdemaeker (1993): cross-species consistency check (pineapple)*

Chen and De Baerdemaeker [19] performed modal analysis on pineapples, identifying resonant modes that correlated with flesh firmness. Their measured frequencies ($\sim$ 100 Hz to 300 Hz) are consistent with the general prediction of the shell model when pineapple-appropriate geometry and rind stiffness are used. This cross-species consistency supports the suggestion that the equivalent-sphere shell framework may generalise beyond watermelons, although a rigorous validation for each species would require purpose-designed experiments.

4.2. *Effect of geometric uncertainty*

In practice, fruit geometry is measured approximately—for example, with callipers, from a photograph, or by mass–density inference. We quantify the effect of geometric measurement error on the inverted modulus through a Monte Carlo simulation. The true tap-tone frequency f_2 is computed from the canonical ripe parameters ($E_{\text{rind}} = 50\,\text{MPa}$, yielding $f_2 = 89.9\,\text{Hz}$), and then the inversion is performed 10 000 times with each geometric parameter (a, c, h) independently perturbed by a uniform $\pm 10\,\%$ error.

Table 9 summarises the results. The inverted modulus is unbiased ($\overline{E}_{\text{inv}} = 50.3\,\text{MPa}$) but exhibits a standard deviation of 6.8 MPa (CoV = 13.5 %), with a 95 % confidence interval of [38.5 MPa, 64.1 MPa]. This level of uncertainty is acceptable for ripeness classification: the $\pm 10\,\%$ geometric error maps to a $\pm 14\,\%$ modulus uncertainty, which is smaller than the factor-of-four modulus change between ripeness categories (Table 3).

We also examine the individual sensitivity of f_2 to each geometric parameter. For the canonical ripe watermelon ($f_2 = 89.9\,\text{Hz}$), a $\pm 10\,\%$ perturbation in each parameter yields:

- a: $\Delta f_2 \in [-8.8\,\%, 10.7\,\%]$ — the strongest geometric sensitivity, consistent with $f_2 \propto a^{-3/2}$;
- c: $\Delta f_2 \in [-4.5\,\%, 5.2\,\%]$;
- h: $\Delta f_2 \in [-4.6\,\%, 4.4\,\%]$;
- ρ_f: $\Delta f_2 \in [-3.8\,\%, 4.3\,\%]$;

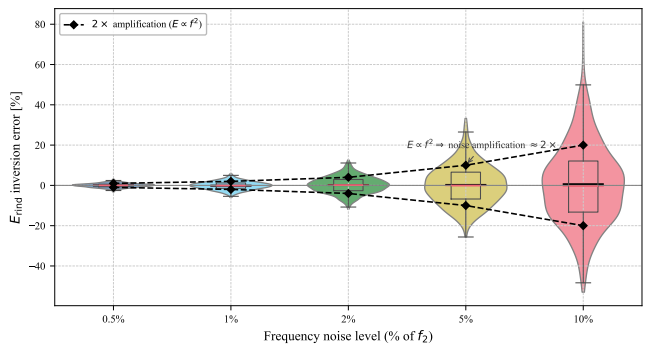

Figure 6: Effect of frequency measurement noise on inverted rind modulus, assessed by 1 000 Monte Carlo trials at each noise level. The condition number of the inversion with respect to f is approximately 2 (since $E \propto f_2^2$, the condition number with respect to f^2 is unity): 2 % frequency noise yields $\approx$ 4 % modulus error. Violin plots show the full distribution; box plots indicate interquartile range.

- ρ_w: $\Delta f_2 \in [-0.9\,\%, 0.9\,\%]$ — negligible.

The semi-major axis a dominates, consistent with the Sobol analysis (§3.2). In a practical measurement scenario, even a crude estimate of the equatorial diameter (obtainable from a single photograph) reduces the dominant source of geometric uncertainty.

5. Discussion

5.1. Why the tap-tone drops with ripening

The monotonic decrease in tap-tone frequency during ripening has a straightforward physical explanation. As the fruit ripens, enzymatic degradation of pectin in the rind cell walls reduces the effective Young's modulus E_{rind}. Since $f_2 \propto \sqrt{E_{\text{rind}}}$ (Eq. 8), the frequency falls. For the canonical Crimson Sweet geometry, f_2 drops from 203 Hz (unripe, $E = 200$ MPa) to 46 Hz (overripe, $E = 15$ MPa)—a factor of 4.4 reduction. This is precisely the information that experienced shoppers extract unconsciously: a lower pitch signals a softer, riper melon.

The frequency shift is not driven by changes in fluid density or fruit geometry, both of which vary by less than 5 % over the ripening window. The Sobol analysis (§3.2) confirms that E_{rind} is the dominant parameter, contributing 54 % of the total variance ($S_T = 0.54$), whereas flesh density and turgor pressure contribute less than 0.1 % each.

Step 1: Tap ⟶ Step 2: Record ⟶ Step 3: Extract f_2 ⟶ Step 4: Input geometry ⟶ Step 5: Invert → E_{rind}

Knuckle / impactor → Microphone / accelerometer → FFT peak detection → (a, c, h) from callipers or photo → Eq. (10)

Figure 7: Envisaged practical workflow for acoustic rind stiffness estimation. The closed-form inversion requires only the tap-tone frequency f_2 and approximate fruit geometry. The subsequent mapping from E_{rind} to a ripeness classification (dashed) requires cultivar-specific calibration data not developed in this study.

The secondary contribution of fruit size ($S_T = 0.38$) is relevant across cultivars but approximately constant during ripening of a single fruit.

The quality factor also decreases during ripening—from $Q = 12.5$ (unripe) to $Q = 5.0$ (overripe)—as the loss tangent increases from 0.08 to 0.20 with pectin degradation. In practice, this means that an overripe melon produces a duller, more heavily damped thump, consistent with the subjective experience of a 'mushy' sound.

5.2. Practical workflow and measurement considerations

Figure 7 illustrates the envisaged measurement workflow. The procedure involves five steps: (1) tap the fruit with a knuckle or small impactor; (2) record the acoustic response with a microphone (or accelerometer attached to the rind); (3) extract the dominant spectral peak f_2 via FFT; (4) input the fruit geometry (a, c, h), obtainable from callipers or a photograph; and (5) apply the closed-form inversion (Eq. 10) to estimate E_{rind}.

Several practical measurement conditions merit discussion:

- **Support conditions.** The fruit should rest on a soft support (e.g. foam pad) to approximate free boundary conditions. A hard surface introduces contact stiffness that may shift or split the $n = 2$ resonance.

- **Tap location.** The $n = 2$ flexural mode has maximum displacement at the equator; tapping near the equatorial plane maximises the modal excitation. Tapping at the poles preferentially excites axisymmetric ($n = 0$) breathing modes at much higher frequencies.

- **Microphone or sensor position.** The microphone should be placed near the fruit surface (within a few centimetres) to maximise the signal-to-noise ratio. An accelerometer bonded to the rind provides a stronger signal but requires contact.

- **Ambient noise.** Outdoor or market environments may mask the tap-tone signal; a brief time-windowed acquisition ($\sim$ 50 ms) centred on the tap event can reject most background noise.

5.3. Limitations

Several simplifications deserve acknowledgement:

- **Equivalent-sphere approximation.** The present model captures geometry through the equivalent-volume sphere radius $R_{\mathrm{eq}} = (a^2c)^{1/3}$ and does not include explicit curvature corrections for oblate deformation. All stiffness and inertia expressions (Eqs. 5–7, 2) are those of a spherical shell. Two watermelons with the same R_{eq} but different aspect ratios ζ are therefore assigned identical frequencies. For watermelons with moderate aspect ratios ($\zeta \approx 0.7$–0.9), this approximation introduces an error of unquantified magnitude in the predicted tap-tone frequency. The dominant physics is expected to be controlled by the shell's average radius rather than its local curvature, but bounding this error rigorously would require comparison to a Ritz variational model with explicit ζ-dependent basis functions (e.g. Legendre polynomials in oblate spheroidal coordinates), which is identified as future work.

- **Isotropic rind.** The watermelon rind is a biological composite with fibre orientation that may induce anisotropy. We treat it as isotropic; orthotropic extensions are straightforward but require additional material characterisation.

- **Uniform thickness.** Real rinds are thicker near the blossom end. Thickness variation could be incorporated via perturbation methods but is unlikely to shift f_2 by more than $\sim$ 10 %, based on the ± 10 % sensitivity analysis in §4.2 ($\Delta f_2 \in [-4.6\,\%, 4.4\,\%]$ for ± 10 % uniform thickness perturbation).

- **Flesh as inviscid fluid.** Watermelon flesh has finite shear strength. At the frequencies of interest ($\sim 100\,\mathrm{Hz}$), the flesh is well within the fluid-like regime ($\omega\tau \ll 1$, where τ is the viscoelastic relaxation time), so this approximation is reasonable.

- **Turgor pressure.** We neglect internal turgor pressure effects on the eigenfrequency in the inversion. The Sobol analysis confirms that P_{int} contributes less than 0.1 % of the total variance in f_2, validating this simplification. For very unripe fruit with intact cell walls, turgor may contribute a correction of order 10 %, but this falls within the geometric uncertainty band.

- **Experimental validation.** The model predictions are compared with published data (§4.1) but have not yet been validated against purpose-designed experiments with controlled geometry and material characterisation. Such validation is a priority for future work.

- **Frequency-to-ripeness chain.** The model establishes the link from tap-tone frequency to effective rind modulus. The further link from modulus to biological ripeness (eating quality, sugar content, texture) requires cultivar-specific calibration against destructive reference measurements, which has not been performed here. The model therefore delivers a *stiffness parameter*, not a ripeness classification.

5.4. Practical implications

The closed-form inversion (Eq. 10) is computationally trivial and could be implemented in a smartphone application. A microphone records the tap-tone, a fast Fourier transform extracts f_2, and the inversion formula returns E_{rind} given approximate fruit dimensions (obtainable from a photograph). The dimensionless geometric invariant Π_{ripe} then provides an approximate cultivar-independent calibration constant that maps the result onto a modulus scale, although the 5.4 % spread across cultivars (Table 6) means that cultivar-specific calibration would be preferable in practice. We reiterate that Π_{ripe} collapses geometries, not ripeness stages; its utility as a ripeness predictor has not been biologically validated.

The Monte Carlo uncertainty analysis (§4.2) demonstrates that even with ±10 % geometric measurement error, the inverted modulus has a coefficient of variation of only 13.5 % (95 % CI: [38.5 MPa, 64.1 MPa] for a true value of 50 MPa). This uncertainty is well within the factor-of-four modulus difference between ripeness categories (Table 3), suggesting that *stiffness discrimination* may be achievable pending cultivar-specific calibration, even with imprecise geometry.

This approach offers several advantages over existing non-destructive methods (near-infrared spectroscopy, impact testing with accelerometers): it requires no specialised hardware, is non-contact (or nearly so—one need only tap), and provides a physics-based rather than purely statistical estimate of internal quality. The Sobol analysis further shows that the parameters most difficult to measure *in situ* (flesh density, Poisson's ratio, turgor pressure) are precisely those to which the tap-tone is least sensitive, a fortuitous alignment that makes the method practically robust.

5.5. Generality of the shell–fluid framework

The model is not fruit-specific. Any fluid-filled biological structure that can be approximated as a thin shell falls within its scope. Candidate applications include other cucurbits (cantaloupe, honeydew, pumpkin), drupes (coconut), and even non-biological systems such as pressurised tanks and sporting goods. The multi-cultivar comparison (§3.4) demonstrates that cultivar-specific geometry is easily accommodated by adjusting three parameters (a, c, h), and the dimensionless geometric invariant Π_{ripe} provides a natural framework for cross-species calibration.

The collapse of Π_{ripe} to within 3.3 % across four geometrically distinct watermelon cultivars (Table 6) suggests that extension to other fruit species requires only measurement of the species-specific shell geometry (a, c, h), which determines R_{eq} and thereby the frequency scaling.

Connection to biomedical vibroacoustics.. The shell–fluid framework used here is structurally identical to models applied to the human abdomen [21], illustrating that classical shell theory provides a common analytical basis for biological fluid-filled structures across very different material scales.

6. Conclusions

We have developed a mechanics-based model for the acoustic 'thump test" used to assess watermelon ripeness. The model delivers an estimate of the *effective rind stiffness parameter*; the subsequent inference from stiffness to biological ripeness requires cultivar-specific calibration that is beyond the scope of this work. The principal findings are:

1. The watermelon is modelled as a fluid-filled viscoelastic shell using an equivalent-sphere approximation: the oblate geometry is captured through $R_{\mathrm{eq}} = (a^2 c)^{1/3}$, and the eigenfrequency is derived from classical thin-shell theory for a sphere of that radius. The dominant tap-tone corresponds to the $n = 2$ flexural mode, identifiable as the lowest spectral peak in the impulse response.
2. The squared frequency f_2^2 is linear in the rind elastic modulus E_{rind} when turgor pressure is negligible. This linearity enables a closed-form inversion (Eq. 10) from a single tap-tone measurement to an estimate of the effective rind stiffness. The condition number of this inversion is unity with respect to f^2 and approximately 2 with respect to f.
3. Sobol sensitivity analysis confirms that E_{rind} dominates the frequency response ($S_T = 0.54 \pm 0.05$), with fruit size as the second most influential parameter.
4. As a secondary result, the dimensionless group Π_{ripe} (Eq. 11) collapses different cultivar geometries onto a narrow band ($\Pi_{\mathrm{ripe}} = 0.062 \pm 0.002$ across four cultivars), although its biological validity as a ripeness index across cultivars has not been established.
5. Model predictions are benchmarked against published resonance measurements on watermelons [1] and are consistent with frequency–stiffness correlations reported for other fruits [7, 19]. These comparisons are consistency checks against literature data, not controlled experimental validation.

The equivalent-sphere approximation is adequate for the inversion application because the goal is to recover E_{rind} from f_2 at known geometry; since a, c, and h are measured independently, R_{eq} is determined and the inversion is well-posed. However,

the model has zero sensitivity to aspect ratio at constant R_{eq}: it cannot distinguish a sphere from an oblate spheroid of equal volume. A Ritz variational model with explicit ζ-dependent basis functions would be needed to capture genuine curvature effects and could improve accuracy for strongly oblate cultivars ($\zeta < 0.6$).

Future work will pursue: (i) experimental validation on instrumented watermelons with destructive reference measurements to establish the modulus–ripeness calibration chain; (ii) extension of the model to include explicit oblate curvature corrections via a Ritz variational formulation; (iii) application to other fruits (cantaloupe, coconut, pumpkin); and (iv) development of a smartphone-based measurement tool.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository at `https://github.com/JonathanMace/brownnote`.

CRediT authorship contribution statement

Jonathan Mace: Conceptualization, Methodology, Software, Supervision, Writing – Original Draft, Writing – Review & Editing. **Brian R. Mace**: Writing – Review & Editing, Validation.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GitHub Copilot (powered by Claude, Anthropic) in order to assist with analytical derivation verification, numerical code development, and manuscript drafting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 2: Canonical parameters for the Crimson Sweet watermelon at optimal ripeness. Geometry (a, c, h) and densities are representative values drawn from Sadrnia et al. [24] and De Belie et al. [7]. The rind modulus is an assumed mid-range value for ripe fruit based on the ranges reported in those sources. Poisson's ratio and turgor pressure are assumed from the literature range for soft biological tissues. The flesh bulk modulus is taken as a standard value for water-like media. The loss tangent is representative of ripe rind tissue.

Parameter	Symbol	Value	Unit	Provenance
Semi-major axis	a	0.158	m	Lit.[a]
Semi-minor axis	c	0.123	m	Lit.[a]
Aspect ratio	$\zeta = c/a$	0.78	—	Derived
Equivalent radius	R_{eq}	0.145	m	Derived
Rind thickness	h	15	mm	Lit.[a]
Rind modulus (ripe)	E_{rind}	50	MPa	Assumed[b]
Poisson's ratio	ν	0.40	—	Assumed[c]
Rind density	ρ_w	1050	kg/m^3	Lit.[a]
Flesh density	ρ_f	950	kg/m^3	Lit.[a]
Flesh bulk modulus	K_f	2.2	GPa	Standard[d]
Turgor pressure	P_{int}	500	Pa	Assumed[c]
Loss tangent	η	0.15	—	Assumed[c]

[a] Measured from literature [24, 7].

[b] Assumed mid-range value for ripe fruit from reported literature ranges.

[c] Assumed from literature range for soft biological tissues.

[d] Standard value for water-like media.

Table 3: Forward model predictions of the $n = 2$ flexural mode frequency f_2 for the canonical Crimson Sweet watermelon at four illustrative ripeness stages. The rind modulus E_{rind}, rind thickness h, loss tangent η, and flesh density ρ_f vary with ripening; all other parameters are held at their canonical values. *Data source:* All values are **constructed scenarios** based on ranges reported in the literature [24, 7]; no stage-resolved measurements from a single cultivar study are available. See text for details.

Stage	E_{rind} [MPa]	h [mm]	η	f_2 [Hz]	Q	f_{d} [Hz]
Unripe	200	20	0.08	203.2	12.5	203.0
Turning	120	17	0.11	146.7	9.1	146.5
Ripe	50	15	0.15	89.9	6.7	89.6
Overripe	15	13	0.20	46.4	5.0	46.2

Table 4: Input parameter ranges for the Sobol sensitivity analysis. Ranges span the full variability expected across commercial watermelon cultivars and ripeness stages.

Parameter	Symbol	Lower bound	Upper bound
Rind modulus	E_{rind}	10 MPa	250 MPa
Rind thickness	h	10 mm	25 mm
Semi-major axis	a	0.090 m	0.220 m
Semi-minor axis	c	0.080 m	0.160 m
Rind density	ρ_w	1000 kg/m^3	1100 kg/m^3
Flesh density	ρ_f	920 kg/m^3	990 kg/m^3
Poisson's ratio	ν	0.35	0.45
Turgor pressure	P_{int}	100 Pa	800 Pa
Loss tangent	η	0.05	0.25

Table 5: Sobol sensitivity indices for the tap-tone frequency f_2. Parameters are ranked by total-order index S_T. Confidence intervals (95%, from bootstrap resampling) are shown as ± values. The rind modulus E_{rind} dominates, contributing over half the total variance.

Parameter	S_1	S_T
E_{rind}	0.494 ± 0.04	0.542 ± 0.05
a	0.331 ± 0.03	0.375 ± 0.04
h	0.072 ± 0.02	0.085 ± 0.02
c	0.047 ± 0.02	0.059 ± 0.02
ν	0.001 ± 0.01	0.001 ± 0.01
ρ_f	< 0.001	< 0.001
ρ_w	< 0.001	< 0.001
P_{int}	< 0.001	< 0.001
η	< 0.001	< 0.001

Table 6: Dimensionless geometric invariant Π_{ripe} for four cultivars evaluated at each cultivar's canonical geometry (fixed a, c, h) with E_{rind} set to four illustrative modulus levels (column headers, in MPa). The near-constant value per cultivar ($\overline{\Pi}_{\text{ripe}} = 0.062$, CoV = 3.3 %) confirms that E_{rind} cancels by construction at fixed geometry. The residual spread across cultivars reflects differences in shell geometry.

Cultivar	E=200	E=120	E=50	E=15 [MPa]
Crimson Sweet	0.0601	0.0601	0.0601	0.0601
Sugar Baby	0.0655	0.0655	0.0654	0.0655
Charleston Gray	0.0621	0.0620	0.0620	0.0620
Seedless (Tri-X 313)	0.0607	0.0607	0.0607	0.0607

Table 7: Multi-cultivar comparison of predicted tap-tone frequency f_2 at the canonical ripe modulus $E_{\text{rind}} = 50\,\text{MPa}$. All four cultivars use $\nu = 0.40$, $\rho_w = 1050\,\text{kg/m}^3$, $\rho_f = 950\,\text{kg/m}^3$. Cultivar geometries are representative values assembled from published dimensions [24, 7]; they do not derive from a single controlled cultivar trial.

Cultivar	a [m]	c [m]	h [mm]	R_{eq} [m]	f_2 [Hz]
Crimson Sweet	0.158	0.123	15	0.145	89.9
Sugar Baby	0.100	0.095	12	0.098	143.4
Charleston Gray	0.200	0.110	18	0.164	82.0
Seedless (Tri-X 313)	0.140	0.120	14	0.133	99.1

Table 8: Comparison of model predictions with the frequency ranges reported by Yamamoto et al. [1]. Model geometry is chosen to match reported fruit sizes (*measured from literature*); material properties are set to canonical ripe ($E = 50\,\text{MPa}$) or unripe ($E = 150\,\text{MPa}$) values (*assumed from literature range*). This is a consistency check, not a point-by-point validation.

Description	a [m]	c [m]	h [mm]	E [MPa]	f_2^{model} [Hz]	Yamamoto range [Hz]
Small, ripe	0.110	0.100	13	50	132	
Medium, ripe	0.140	0.120	15	50	102	80–120
Large, ripe	0.170	0.140	17	50	83	
Small, unripe	0.110	0.100	15	150	245	
Medium, unripe	0.140	0.120	18	150	192	120–180
Large, unripe	0.170	0.140	20	150	155	

Table 9: Effect of ±10 % geometric measurement uncertainty on the inverted rind modulus, evaluated by Monte Carlo simulation ($n = 10{,}000$) for two ripeness stages.

Stage	E_{true} [MPa]	$\overline{E}_{\text{inv}}$ [MPa]	σ_E [MPa]	CoV [%]	95 % CI [MPa]
Ripe	50	50.3	6.8	13.5	[38.5, 64.1]
Unripe	200	201.3	27.7	13.7	—